

* Artificial Intelligence

technique to mimic human intelligence

AI is a broader concept of making machines intelligent while ML is a subset of AI that enables machines to learn from data and improve automatically.

* NLP (Natural Language Processing)

NLP means

NLU - Natural language understanding

NLG - Natural language generation

NLP handles text type of data that comes from social media, comments, pdfs, movies etc.

Machine learning handles tabular data -- excel, csv file.

NLP is a field of Artificial Intelligence (AI) that helps computers understand, interpret and respond to human language (text or speech)

* Tokenization

The words are called tokens.

Break a complex sentence into words

① word_tokenize()

Split texts into words + punctuation

Eg. "Hello! How are you?" $\Rightarrow$ ['Hello', '!', 'How', 'are', 'you', '?']

② `sentence_tokenize()`

Splits text into sentences

Eg: "I love NLP. It is very interesting!"



['I love NLP.', 'It is very interesting!']

③ `blankline_tokenize()`

Splits text into paragraphs based on blank lines.

Eg: "This is paragraph one. \n\n This is paragraph two."



['This is paragraph one.', 'This is paragraph two.']

④ `WhitespaceTokenizer`

Splits text only on spaces ignores punctuation for splitting

Eg: "Hello, NLP World"



['Hello', 'NLP', 'World']

⑤ `wordpunct_tokenize()`

Splits words and punctuation separately.

Eg: "Hello, NLP! Version 3.13"

['Hello', ',', 'NLP', '!', 'Version', '3', '.', '13']

* Types of tokenization.

- ① bigram -- token with 2^{consecutive} words
- ② trigram -- token with 3 consecutive words
- ③ ngram -- token with more than 2 words.

Stemming

Stem the word $\Rightarrow$ gives the root form of the word.

playing, played $\Rightarrow$ play

$\Downarrow$
(root word)

Stemming will do suffix stripping not prefix.

Stemming will give root word.

Stemming divided into 3 parts:

- ① PorterStemmer
- ② LancasterStemmer
- ③ SnowballStemmer

① PorterStemmer:

Balanced stemmer. Not too strong, not too weak

② LancasterStemmer:

Cuts words very strongly. Sometimes removes too much.

③ SnowballStemmer

More accurate than PorterStemmer. Supports many languages which is not supported in case of PorterStemmer and LancasterStemmer.

Lemmatization

Lemmatization will give complete word

playing $\rightarrow$ stemming $\rightarrow$ play
lemmatizing $\rightarrow$ playing.

Lemmatization is used to preserve the correct meaning of words by converting them into their dictionary form even when stemming fails.

children $\rightarrow$ stemming $\rightarrow$ children

children $\rightarrow$ lemmatizing $\rightarrow$ child.

Stop words:

In every language we are using common words like "the", "is", "in". These are filtered during text cleaning.

By removing the stopwords we will get best accuracy.

POS -- part of speech like adverb, verb, noun etc.

NER -- Named entity recognition

Identifies and classifies important real-world names in the text.

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LOCATION India

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PERCENT 10%.